

Milestone 3

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Treatment and Covariates

Below are the TableOne results with standardized mean differences for each row in the data set, which corresponds to one pair of back to back games. The treatment group includes only pairs of back to backs in which one of the games was missed due to load management, and the control group includes only pairs of back to backs in which both games were played. A game missed due to load management is defined as a game in which the player did not play but played in the games both before and after. This is the criteria that was chosen because teams give a variety of injury designations in these cases such as tightness or soreness rather than outright saying that it's exclusively due to rest even when it is the case that the player would have been able to play if not for the decision to manage their workload, so defining games missed due to load management this way was deemed a superior option to defining them in terms of the team's injury designation.

The covariates shown here are those used to fit the propensity score model. These include:

distance_average: The average flight distance per game played in the back to back pair. Research suggests that air travel has a negative influence on player health⁴.

Rolling_Average_MP: The five-game rolling average of minutes played prior to the back-to-back set. Research shows that increases in minutes per game in the preceding window of games have a significant effect on injuries, with that of a five game window being the most significant³.

Age: The player's age as of February 1st in the given season.

MPG: The player's minutes per game in a given season.

Games_Missed_Frequency_Career: The player's frequency of games missed throughout their career before the season in question. This is included to help address the confounding nature of more injury prone players being rested in back to backs.

Positional Indicators (PG, SG, SF, PF, C): Indicators for which position the player played. This is based on the season long depth charts not individual games.

Pandemic-Influenced Season Indicators (Is_2019_20, Is_2020_21, Is_2021_22): Indicators for whether the game was in the 2019-20, 2020-21, or 2021-22 seasons. These indicators are included because each of these seasons and/or the length of the offseason leading up to them were influenced by the COVID-19 pandemic.

Schedule Density Indicators (Is_3rd_Game_In_4_Days, Is_4th_Game_In_5_Days, Is_5th_Game_In_7_Days): Indicators for the frequency of games as of the second game of the back to back.

##		Stratified by treatment	
##		0	1
##	n	17721	1316
##	distance_average (mean (SD))	542.29 (333.14)	577.17 (538.01)
##	Rolling_Avg_MP (mean (SD))	27.95 (7.87)	25.97 (9.02)
##	Age (mean (SD))	26.17 (4.04)	28.11 (4.37)
##	MPG (mean (SD))	29.69 (4.89)	30.17 (4.71)
##	Games_Missed_Frequency_Career (mean (SD))	0.09 (0.10)	0.11 (0.09)

##	PG (mean (SD))	0.19 (0.40)	0.22 (0.42)
##	SG (mean (SD))	0.20 (0.40)	0.17 (0.38)
##	SF (mean (SD))	0.21 (0.41)	0.17 (0.38)
##	PF (mean (SD))	0.20 (0.40)	0.23 (0.42)
##	C (mean (SD))	0.20 (0.40)	0.20 (0.40)
##	Is_2019_20 (mean (SD))	0.06 (0.24)	0.07 (0.26)
##	Is_2020_21 (mean (SD))	0.09 (0.29)	0.14 (0.34)
##	Is_2021_22 (mean (SD))	0.08 (0.27)	0.14 (0.34)
##	Is_3rd_Game_in_4_Days (mean (SD))	0.68 (0.47)	0.76 (0.43)
##	Is_4th_Game_in_5_Days (mean (SD))	0.04 (0.20)	0.03 (0.18)
##	Is_5th_Game_in_7_Days (mean (SD))	0.12 (0.32)	0.16 (0.36)
##		Stratified by treatment	
##		SMD	
##	n		
##	distance_average (mean (SD))	0.078	
##	Rolling_Avg_MP (mean (SD))	0.234	
##	Age (mean (SD))	0.462	
##	MPG (mean (SD))	0.100	
##	Games_Missed_Frequency_Career (mean (SD))	0.182	
##	PG (mean (SD))	0.075	
##	SG (mean (SD))	0.083	
##	SF (mean (SD))	0.085	
##	PF (mean (SD))	0.075	
##	C (mean (SD))	0.011	
##	Is_2019_20 (mean (SD))	0.030	
##	Is_2020_21 (mean (SD))	0.142	
##	Is_2021_22 (mean (SD))	0.177	
##	Is_3rd_Game_in_4_Days (mean (SD))	0.168	
##	Is_4th_Game_in_5_Days (mean (SD))	0.031	
##	Is_5th_Game_in_7_Days (mean (SD))	0.118	

Initial investigation of covariates showed a couple of concerns regarding data leakage if they were used to fit a propensity score model and a few changes were made. The travel data was almost always empty for cases when a player didn't play a game, suggesting either that the player did not travel with the team or that the data was incorrect. In either case, this would be data leakage because any of the variables related to travel being 0 often indicated when a player sat a game, which is the outcome. Originally the intent was to include whether the player travelled to each game of the back to back and the sum of the flight distances, but now the indicators were removed and the average flight distance per game played was used instead of the sum of distances. This average should be more informative because if a player had to endure a long flight before one game of a back to back their team may have been more likely to rest them in the other game and also possibly more likely to get injured, and as expected the mean in the treatment group is slightly larger. A similar problem arose with the five game rolling average minutes played because this average includes a 0 for games not played, so it was significantly lower for the back to back pairs as a result when it was initially created including the games in the back to back set in the rolling average calculation. This was revised to avoid data leakage by using the five-game rolling average of minutes played prior to the back to back set, so that it would not be systematically lower for cases when a player was rested in half of the back to back. In addition, when the schedule density indicators (Is_3rd_Game_In_4_Days, Is_4th_Game_In_5_Days, Is_5th_Game_In_7_Days) were originally calculated, they were almost always 0 for cases when the player didn't play half of the back to back because it will never be, for example, the third game played in four nights if a player sits one of them. So these were recalculated to represent, for example, whether it would've been the third game in four nights had the player played in both games of the back to back, which is a much more meaningful covariate in the context of this study and has a much more reasonable standardized mean difference. Finally, the total games played in the season was also removed from the propensity score covariates because players who miss back to backs are playing fewer games as a result so it would be data leakage to

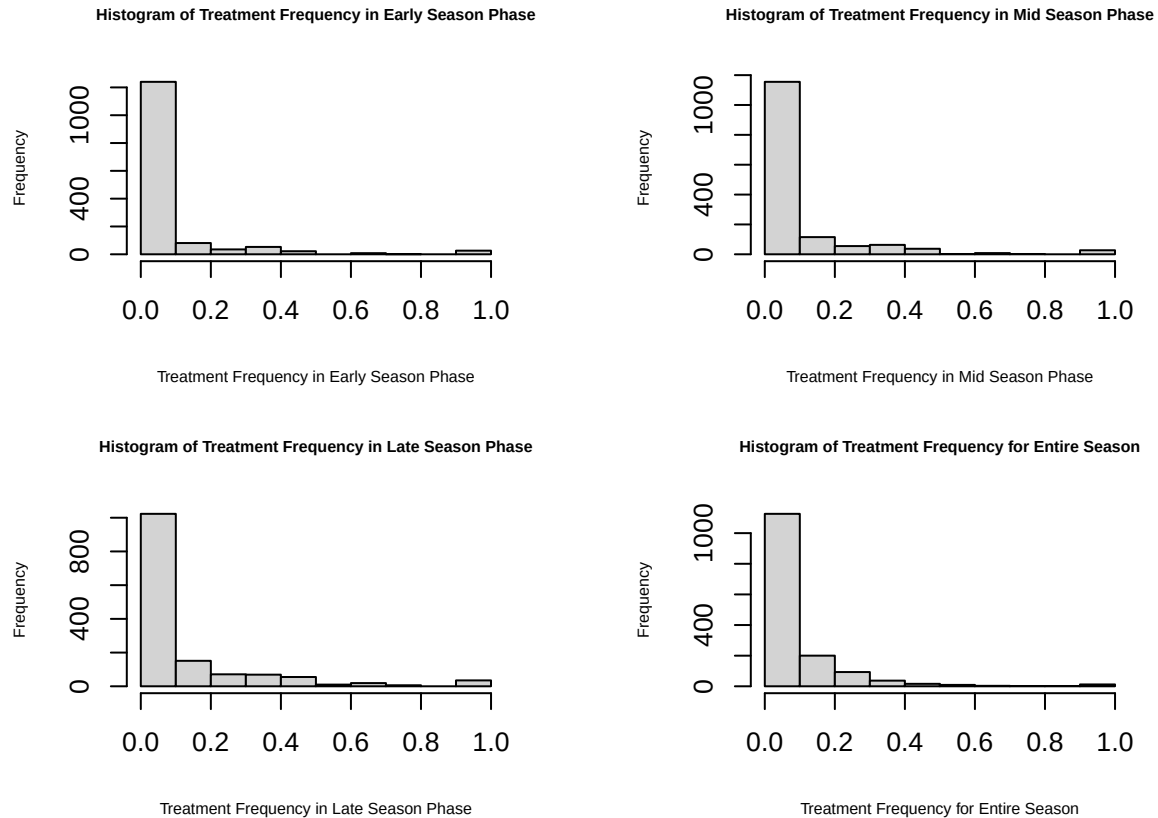
include this in the propensity score model. However, total games played, as well as the other season-level covariates (Age, MPG, career games missed frequency, pandemic-influenced season indicators) will also be included in the marginal structural model that uses the propensity scores to predict the season-level injury outcomes.

The table contains the covariates after these changes were made, and we no longer see any unreasonably large standardized mean differences as a result.

Outcomes

The outcomes are injury frequency, which is the number of games in which an injury occurred divided by the number of games played for a player in a given season, and games missed frequency, which is the number of games that a player missed due to injury divided by the number of games played by their team in the given season. Importantly, games missed due to injury exclude those missed for load management. This was enforced in order to avoid data leakage in the sense that a back to back game missed, which is the treatment, would directly contribute to the games missed frequency.

These two outcomes were examined for four different season phases. These are early-season, mid-season, late-season, and the entire season. The first third of a team's games is the early-season phase, the next third is the mid-season phase, and the final third is the late-season phase. These season phases are used to determine if there is significantly different outcomes based on when treatment occurs throughout the season and preserving the time-varying nature of treatment without analyzing very large numbers of possible treatment sequences. The entire season phase is included along with the early, mid, and late season phases to allow for more general analysis. For each of the two outcomes within each of the four season phases, there are three strata of treatment. The first is the case where there was no treatment in the given phase. The second and third represent a low nonzero treatment frequency in a given phase and a high nonzero treatment frequency in a given phase respectively. These were determined by splitting the frequencies to treatment at a given phase at the median amongst frequencies that are nonzero. This procedure for determining the stratification of treatment was performed in order to have meaningful distinctions between frequencies of treatment given the distribution of treatment frequency is heavily skewed right for all four season phases as shown in the histograms below. The benefit of the way the treatment frequencies were stratified is that the frequency of zero represents its own category since it is by far the largest and there is still a distinction between a low nonzero frequency of treatment and a high nonzero frequency of treatment.



Unadjusted Mean Outcomes

Below are the mean, standard error and 95% confidence interval for each of the three strata (Q1 = no treatment, Q2 = low treatment frequency, Q3 = high treatment frequency) in each of the four categories for each of the two outcome variables. Standard errors are calculated using bootstrap resampling.

```
## # A tibble: 24 x 7
```

##	Outcome	Stratum	Phase	Mean	SE	CI_Lower	CI_Upper
##	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 Injury_Frequency	Q1	Early	0.0299	0.000894	0.0282	0.0316
##	2 Injury_Frequency	Q2	Early	0.0384	0.00342	0.0316	0.0451
##	3 Injury_Frequency	Q3	Early	0.0467	0.00343	0.0401	0.0535
##	4 Injury_Frequency	Q1	Mid	0.0291	0.000908	0.0274	0.0309
##	5 Injury_Frequency	Q2	Mid	0.0378	0.00250	0.0330	0.0428
##	6 Injury_Frequency	Q3	Mid	0.0473	0.00316	0.0413	0.0533
##	7 Injury_Frequency	Q1	Late	0.0280	0.000904	0.0263	0.0299
##	8 Injury_Frequency	Q2	Late	0.0324	0.00204	0.0285	0.0364
##	9 Injury_Frequency	Q3	Late	0.0443	0.00248	0.0394	0.0490
##	10 Injury_Frequency	Q1	Total	0.0269	0.00109	0.0248	0.0291
##	11 Injury_Frequency	Q2	Total	0.0315	0.00146	0.0287	0.0345
##	12 Injury_Frequency	Q3	Total	0.0465	0.00207	0.0422	0.0503
##	13 Games_Missed_Frequency	Q1	Early	0.0993	0.00341	0.0926	0.106
##	14 Games_Missed_Frequency	Q2	Early	0.107	0.0119	0.0843	0.130
##	15 Games_Missed_Frequency	Q3	Early	0.129	0.0119	0.107	0.154
##	16 Games_Missed_Frequency	Q1	Mid	0.0945	0.00353	0.0875	0.101
##	17 Games_Missed_Frequency	Q2	Mid	0.0921	0.00679	0.0790	0.107

## 18 Games_Missed_Frequency Q3	Mid	0.141	0.0117	0.118	0.165
## 19 Games_Missed_Frequency Q1	Late	0.0863	0.00339	0.0802	0.0933
## 20 Games_Missed_Frequency Q2	Late	0.0832	0.00607	0.0710	0.0956
## 21 Games_Missed_Frequency Q3	Late	0.137	0.00964	0.118	0.156
## 22 Games_Missed_Frequency Q1	Total	0.0954	0.00444	0.0867	0.104
## 23 Games_Missed_Frequency Q2	Total	0.0887	0.00542	0.0782	0.0995
## 24 Games_Missed_Frequency Q3	Total	0.137	0.00754	0.122	0.151

Models

The models that were fit for this project are a propensity score model using all of the covariates in the table in the Treatment and Covariates section and a marginal structural model (MSM) that estimates the mean of one of the outcomes in a given season phase using the treatment along with the season-level covariates and weighted using the inverse probability weights. Since the propensity scores are calculated for each game and the outcomes of the MSM are over the course of the entire season, the weights of the MSM are calculated using the sum of propensity scores for a given player within the given season phase. The propensity score model will only be fit once because it is only used to predict the binary treatment of being held out of one game of a back to back. The MSM was fit for eight different combinations of outcome and season phase. The two outcomes are injury frequency and games missed frequency within each of the season phases as described in the Outcomes section. The model estimates the mean for each stratum for each of these outcomes and season phases.

Missing covariate values are not prevalent because the data sources are very reliable and any inconsistencies have been eliminated in the process of transforming the data. However, there are cases in which a player did not play in any back to backs in a given season phase (early mid, or late) in a given season. In this case the player in the given season is not considered in the model for that season phase. No variable selection methods were used to fit either the propensity score model or the MSM.

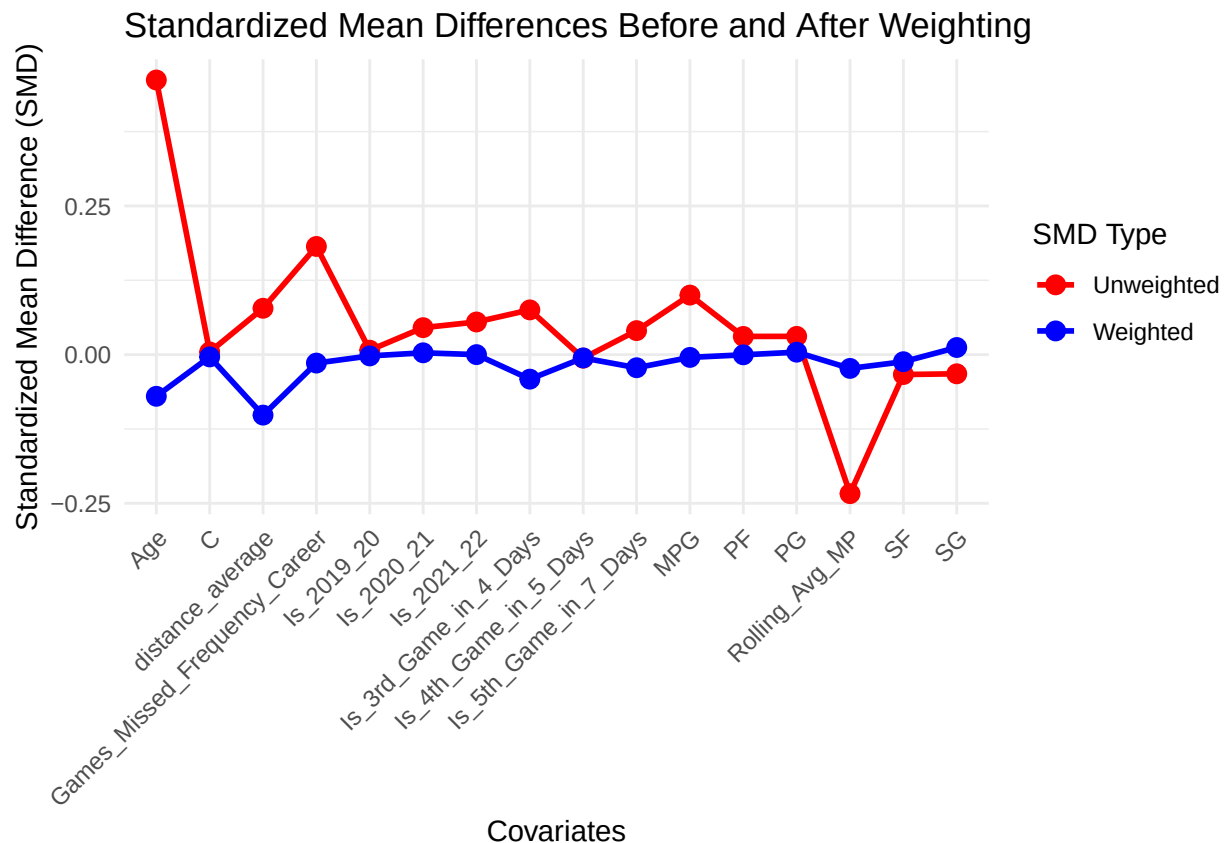
Results

Below is the summary of the propensity score model:

```
##
## Call:
## glm(formula = propensity_formula, family = binomial(link = "logit"),
##      data = b2b_of_interest)
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -7.290e+00  3.005e-01 -24.263 < 2e-16 ***
## distance_average  3.057e-04  7.847e-05   3.895 9.82e-05 ***
## Rolling_Avg_MP  -5.060e-02  3.840e-03 -13.175 < 2e-16 ***
## Age            1.142e-01  7.009e-03  16.292 < 2e-16 ***
## MPG            7.305e-02  7.582e-03   9.634 < 2e-16 ***
## Games_Missed_Frequency_Career 1.607e+00  2.921e-01   5.501 3.78e-08 ***
## PG             3.822e-02  9.293e-02   0.411 0.68089
## SG            -1.661e-01  9.749e-02 -1.704 0.08836 .
## SF            -2.432e-01  9.679e-02 -2.513 0.01197 *
## PF             3.065e-02  9.014e-02   0.340 0.73386
## C              NA         NA      NA      NA
## Is_2019_20      3.085e-01  1.164e-01   2.651 0.00802 **
## Is_2020_21      4.898e-01  8.921e-02   5.490 4.02e-08 ***
## Is_2021_22      5.547e-01  8.937e-02   6.207 5.42e-10 ***
## Is_3rd_Game_in_4_Days  4.578e-01  7.087e-02   6.459 1.05e-10 ***
## Is_4th_Game_in_5_Days -2.298e-01  1.630e-01 -1.410 0.15859
## Is_5th_Game_in_7_Days  3.490e-01  8.695e-02   4.014 5.97e-05 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 9571.0  on 19036  degrees of freedom
## Residual deviance: 8939.4  on 19021  degrees of freedom
## AIC: 8971.4
##
## Number of Fisher Scoring iterations: 6
```

Below is a plot of the weighted vs unweighted standardized mean differences (SMDs). As anticipated, none of the weighted SMDs are significantly greater than 0.1 in magnitude. Note that these weights are not the same weights as the ones used in the MSM. These are traditional inverse probability weights to align with the game-level covariates, while those used in the MSM are the sum of these weights in a given season phase for a given player to align with the season-level covariates and outcomes of the MSM.



Below are the estimated mean, standard error and 95% confidence interval for each of the three strata (Q1 = no treatment, Q2 = low treatment frequency, Q3 = high treatment frequency) in each of the four categories for each of the two outcome variables using the MSM. Standard errors are calculated using bootstrap resampling.

```
## # A tibble: 24 x 7
##   Quantile Mean      SE CI_Lower CI_Upper Outcome      Phase
##   <chr>    <dbl>    <dbl>    <dbl>    <dbl> <chr>      <chr>
## 1 Q1      0.0279 0.000851 0.0263 0.0296 Injury_Frequency Early
## 2 Q2      0.0357 0.00295 0.0300 0.0414 Injury_Frequency Early
## 3 Q3      0.0451 0.00399 0.0374 0.0532 Injury_Frequency Early
## 4 Q1      0.0908 0.00372 0.0840 0.0984 Games_Missed_Frequency Early
```

##	5	Q2	0.0941	0.0101	0.0754	0.115	Games_Missed_Frequency	Early
##	6	Q3	0.127	0.0140	0.101	0.154	Games_Missed_Frequency	Early
##	7	Q1	0.0262	0.000859	0.0247	0.0280	Injury_Frequency	Mid
##	8	Q2	0.0361	0.00242	0.0315	0.0409	Injury_Frequency	Mid
##	9	Q3	0.0402	0.00307	0.0345	0.0465	Injury_Frequency	Mid
##	10	Q1	0.0793	0.00287	0.0740	0.0850	Games_Missed_Frequency	Mid
##	11	Q2	0.0887	0.00740	0.0751	0.103	Games_Missed_Frequency	Mid
##	12	Q3	0.118	0.0130	0.0942	0.146	Games_Missed_Frequency	Mid
##	13	Q1	0.0250	0.000928	0.0232	0.0268	Injury_Frequency	Late
##	14	Q2	0.0293	0.00176	0.0259	0.0327	Injury_Frequency	Late
##	15	Q3	0.0389	0.00277	0.0335	0.0449	Injury_Frequency	Late
##	16	Q1	0.0729	0.00322	0.0667	0.0793	Games_Missed_Frequency	Late
##	17	Q2	0.0778	0.00623	0.0655	0.0904	Games_Missed_Frequency	Late
##	18	Q3	0.114	0.0104	0.0955	0.135	Games_Missed_Frequency	Late
##	19	Q1	0.0243	0.000907	0.0225	0.0261	Injury_Frequency	Total
##	20	Q2	0.0291	0.00145	0.0264	0.0321	Injury_Frequency	Total
##	21	Q3	0.0412	0.00210	0.0370	0.0453	Injury_Frequency	Total
##	22	Q1	0.0792	0.00353	0.0725	0.0860	Games_Missed_Frequency	Total
##	23	Q2	0.0777	0.00467	0.0683	0.0872	Games_Missed_Frequency	Total
##	24	Q3	0.115	0.00701	0.101	0.129	Games_Missed_Frequency	Total

Below are the differences in means and tests for significance using a two sample Z test between strata for the same outcome and phase.

```
## # A tibble: 24 x 5
##   Outcome                Phase Comparison Difference Significant_0.01
##   <chr>                  <chr> <chr>          <dbl> <lgl>
## 1 Injury_Frequency      Early Q1 vs Q2      -0.00775 FALSE
## 2 Injury_Frequency      Early Q2 vs Q3      -0.00941 FALSE
## 3 Injury_Frequency      Early Q1 vs Q3      -0.0172  TRUE
## 4 Injury_Frequency      Mid   Q1 vs Q2      -0.00984 TRUE
## 5 Injury_Frequency      Mid   Q2 vs Q3      -0.00414 FALSE
## 6 Injury_Frequency      Mid   Q1 vs Q3      -0.0140  TRUE
## 7 Injury_Frequency      Late  Q1 vs Q2      -0.00428 FALSE
## 8 Injury_Frequency      Late  Q2 vs Q3      -0.00958 TRUE
## 9 Injury_Frequency      Late  Q1 vs Q3      -0.0139  TRUE
## 10 Injury_Frequency     Total  Q1 vs Q2      -0.00482 TRUE
## 11 Injury_Frequency     Total  Q2 vs Q3      -0.0121  TRUE
## 12 Injury_Frequency     Total  Q1 vs Q3      -0.0169  TRUE
## 13 Games_Missed_Frequency Early Q1 vs Q2      -0.00333 FALSE
## 14 Games_Missed_Frequency Early Q2 vs Q3      -0.0328  FALSE
## 15 Games_Missed_Frequency Early Q1 vs Q3      -0.0362  FALSE
## 16 Games_Missed_Frequency Mid   Q1 vs Q2      -0.00935 FALSE
## 17 Games_Missed_Frequency Mid   Q2 vs Q3      -0.0295  FALSE
## 18 Games_Missed_Frequency Mid   Q1 vs Q3      -0.0388  TRUE
## 19 Games_Missed_Frequency Late  Q1 vs Q2      -0.00488 FALSE
## 20 Games_Missed_Frequency Late  Q2 vs Q3      -0.0363  TRUE
## 21 Games_Missed_Frequency Late  Q1 vs Q3      -0.0412  TRUE
## 22 Games_Missed_Frequency Total  Q1 vs Q2       0.00153 FALSE
## 23 Games_Missed_Frequency Total  Q2 vs Q3      -0.0370  TRUE
## 24 Games_Missed_Frequency Total  Q1 vs Q3      -0.0355  TRUE
```

Discussion

The results in the previous section do not provide evidence that holding players out of back to back games reduces either injury frequency or frequency of games missed. In fact, depending on the season phase, many of the estimates for injury frequency and frequency of games missed were significantly higher for the groups which received treatment than the group that did not. The differences are particularly striking for the group which had a high frequency of treatment (Q3).

For both injury frequency and frequency of games missed, the estimates are lower for all strata for the late season phase than both the early and mid season phases. One possible explanation for this is that some injury prone players are not included in the late season phase because they were injured in the early or mid season phase and did not play any games in the late season phase as a result. Since this occurs across all strata and not just in treated groups, there is no evidence to suggest that resting a player during one phase of the season yields better injury outcomes than another.

A wide breadth of potential confounding variables that may influence both the decision to hold a player out of half of a back to back and the likelihood and severity of injury are considered in this study. However, it is critical to acknowledge that the techniques used in this study require the assumption that there are no unmeasured confounding variables and that this may be violated. In particular, the fact that the mean estimates using the MSM are often close to their unadjusted counterparts suggests that it is quite possible that there are additional confounding factors. It is a particularly difficult problem because teams are strongly incentivized to sit players that they fear are likely to be injured and this assessment may involve many variables or considerations other than those included in this study.